

Chapter 1

From dg-Shifted Yangians to Spectral Factorization Quantum Groups

1.1 Orientation, status, and first principles

Theorem ?? shows that the open-colour line-side algebra $\mathcal{A}_{\text{line}}^!$ on the chirally Koszul locus carries the dg-shifted Yangian package, whose axioms are the $\text{SC}^{\text{ch},\text{top}}$ -operadic identities decomposed by colour. The spectral Drinfeld strictification obstruction vanishes for all simple Lie algebras (Theorem ??), relating this package to both spectral factorization quantum groups and shifted quantum groups.

There are three languages in play. The physical identifications of Dimofte–Niu–Py [?] are recovered as the naming of operadic data in physical language (Remark ??). The second is the spectral language of factorisation quantum groups, as developed by Latyntsev [?]. The third is the geometric language of shifted quantum groups and critical stable envelopes, as developed in the shifted-Yangian literature and the critical-stable-envelope programme [?].

The guiding problem of this chapter is to put these three languages on one common spine. The core claim is not that a dg-shifted Yangian, viewed through Theorem ?? on the chirally Koszul locus, is *already* a strict factorization quantum group. The correct claim is subtler and stronger: the $\text{SC}^{\text{ch},\text{top}}$ -algebra structure, once expressed through a bar-horizontal strictification of its open-color projection, canonically produces a *spectral quasi-factorization quantum group*, and the passage from quasi to strict is governed by a concrete obstruction complex whose first low-filtration sectors can be computed exactly.

The new results proved here are the following.

1. A bar-horizontal strictification formalism that turns the dg Maurer–Cartan kernel into a flat ordered configuration-space connection.
2. A quasi-factorization theorem: every residue-bounded complete dg-shifted

Yangian equipped with a bar-horizontal strictification canonically yields a spectral quasi-factorization quantum group.

3. A spectral Drinfeld obstruction class controlling strictification, and its complete vanishing for all simple Lie algebras (Theorem ??).
4. The hero: *root multiplicity one*. Every root of a simple Lie algebra has multiplicity one (Theorem ??). This single fact forces the multilinear Lie space controlling each filtration sector to be one-dimensional. A cohomological obstruction valued in a one-dimensional space is a scalar, determined by one equation, which the coproduct rigidity supplies. The explicit BCH coefficient $1/n$ at filtration n (computed by the beta integral $\int_0^1 (1-t)^{n-1} dt = 1/n$ in Theorem ??) identifies the scalar but is irrelevant to the vanishing of the obstruction class: it is the one-dimensionality that does the work.
5. Exact computations in the Cartan, commuting, adjacent-root (triangle), and root-quadrilateral sectors, now seen as the first cases of the all-filtration pattern.
6. A correction and fortification of the naive “shift-invariance” slogan: at mixed-root level the shift does change the local transport coefficients, but only by an explicit transport law that is absorbed at every filtration.

We do not attempt to impose a strict R -matrix directly on the A_∞ object. The operadic reason: the R -matrix is a closed-color datum (from $\text{FM}_2()$), while the A_∞ structure is the open-color E_1 -projection of the $\text{SC}^{\text{ch}, \text{top}}$ -algebra on the chirally Koszul locus (Theorem ??). Imposing the R -matrix on the A_∞ object would conflate colors. Instead, we pass to ordered configuration spaces (the open-color stage), construct a flat connection, read off regularized holonomy, and only then formulate the spectral factorization problem. In the language of the Steinberg presentation (Theorem ??), the flat connection is the spectral connection ∇_z of (iii)(b), and strictification asks whether its monodromy $R(z)$ can be made into a strict factorization R -matrix, the passage from the homotopy-coherent correspondence to its strict shadow.

1.2 Complete filtered spectral setup

Definition 1 (Residue-bounded complete dg-shifted Yangian) *Let Y be a complete separated filtered A_∞ -algebra*

$$Y = F^1 Y \supset F^2 Y \supset \cdots, \quad Y \cong \varprojlim_N Y/F^{N+1} Y,$$

equipped with:

1. a translation family τ_z ,
2. a twisted coproduct Δ_z ,

3. an odd Maurer–Cartan kernel

$$r(z) = \sum_{n \geq 1} r_n z^{-n} \in (Y \hat{\otimes} Y)((z^{-1})),$$

such that

$$m_k(F^{p_1}Y, \dots, F^{p_k}Y) \subset F^{p_1 + \dots + p_k + k - 2}Y,$$

and such that the residues become more convergent with pole order in the sense that

$$r_n(F^pY \hat{\otimes} F^qY) \subset F^{p+q+n}(Y \hat{\otimes} Y).$$

We call such an object a residue-bounded complete dg-shifted Yangian.

The suspension of $r(z)$ to the bar convolution algebra will be denoted by $\tilde{r}(z)$; this is the degree-zero spectral kernel.

Definition 2 (Bar-horizontal strictification) *Let $B(Y) = T^c(sY)$ be the completed reduced bar coalgebra of Y with bar differential b . A bar-horizontal strictification of Y is a complete filtered cochain complex (V, d_V) quasi-isomorphic to $B(Y)$ together with continuous degree-zero endomorphisms*

$$\Omega_{ij}(u) \in \text{End}(V^{\hat{\otimes} n})((u^{-1})) \quad (1 \leq i < j \leq n)$$

such that:

$$\begin{aligned} [d_V, \Omega_{ij}(u)] &= 0, \\ [\Omega_{ij}(u), \Omega_{kl}(v)] &= 0 \quad \text{if } \{i, j\} \cap \{k, l\} = \emptyset, \end{aligned}$$

and for every triple $i < j < k$,

$$[\Omega_{ij}(u), \Omega_{ik}(u+v)] + [\Omega_{ij}(u), \Omega_{jk}(v)] + [\Omega_{ik}(u+v), \Omega_{jk}(v)] = 0.$$

This is the additive spectral precursor of a strict spectral R -matrix. It is the correct object to strictify first.

Remark 3 (Existence of bar-horizontal strictifications) *The existence of a bar-horizontal strictification for HT theories arising from Theorem ?? is established by the explicit Feynman-diagrammatic construction of the KZ connection in Theorem ?? below: the bar-horizontal datum $(V, d_V, \Omega_{ij}(u))$ is produced by the perturbative expansion of the holonomy, with Ω_{ij} given by the tree-level exchange amplitude.*

Theorem 4 (Derived additive KZ connection; Proved here) *Let (V, Ω) be a bar-horizontal strictification. On the ordered configuration space*

$$\text{Conf}_n^{\text{ord}}(\mathbb{A}^1) = \{z_1 < \dots < z_n\}$$

consider the operator

$$\mathbb{D}_n = d_V + d_{\text{dR}} - \sum_{1 \leq i < j \leq n} \Omega_{ij}(z_j - z_i) d(z_j - z_i).$$

Then $\mathbb{D}_n^2 = 0$.

Write

$$A_n := \sum_{i < j} \Omega_{ij}(z_j - z_i) d(z_j - z_i).$$

Since $d_V^2 = d_{\text{dR}}^2 = 0$ and $[d_V, \Omega_{ij}(u)] = 0$, one has

$$(d_V + d_{\text{dR}})^2 = 0, \quad [d_V + d_{\text{dR}}, A_n] = 0,$$

because $d_{\text{dR}}(f(u)du) = f'(u)du \wedge du = 0$. Hence

$$\mathbb{D}_n^2 = A_n \wedge A_n.$$

Disjoint-pair terms vanish by hypothesis. Thus only triples $i < j < k$ contribute. Put

$$u = z_j - z_i, \quad v = z_k - z_j, \quad z_k - z_i = u + v.$$

Then

$$d(z_k - z_i) = du + dv,$$

and every wedge product appearing from the triple $i < j < k$ reduces to a multiple of $du \wedge dv$. The coefficient is precisely

$$[\Omega_{ij}(u), \Omega_{ik}(u+v)] + [\Omega_{ij}(u), \Omega_{jk}(v)] + [\Omega_{ik}(u+v), \Omega_{jk}(v)],$$

which vanishes by the defining spectral Kohno relation. Therefore $\mathbb{D}_n^2 = 0$.

Corollary 5 (Derived Drinfeld–Kohno shadow; Proved here) *If the filtration is pronilpotent, the parallel transport of $\mathbb{D}_n$ is well-defined and yields chain-level pure braid group representations on $V^{\widehat{\otimes} n}$, hence braid-group actions on $H^\bullet(V^{\widehat{\otimes} n})$.*

Theorem 6 (Boundary residue theorem; Proved here) *Assume that each $\Omega_{ij}(u)$ has at most a simple pole at $u = 0$, so that*

$$\Omega_{ij}(u) = \frac{r_{ij}}{u} + \Omega_{ij}^{\text{reg}}(u).$$

Then the connection form of Theorem ?? extends logarithmically to the Fulton–MacPherson compactification $\overline{\text{Conf}}_n^{\text{ord}}(\mathbb{A}^1)$. For the boundary divisor D_S corresponding to collision of a subset $S \subset \{1, \dots, n\}$, its residue is

$$\text{Res}_{D_S}(A_n) = \sum_{i < j \in S} r_{ij}.$$

Near the divisor D_S , choose local coordinates

$$z_i = z_* + t\xi_i \quad (i \in S),$$

with $t = 0$ cutting out the boundary and the ξ_i pairwise distinct. Then for $i, j \in S$,

$$z_j - z_i = t(\xi_j - \xi_i), \quad d(z_j - z_i) = (\xi_j - \xi_i)dt + t d(\xi_j - \xi_i).$$

Hence

$$\begin{aligned}\Omega_{ij}(z_j - z_i) d(z_j - z_i) &= \left(\frac{r_{ij}}{t(\xi_j - \xi_i)} + \text{regular} \right) ((\xi_j - \xi_i)dt + t d(\xi_j - \xi_i)) \\ &= r_{ij} \frac{dt}{t} + r_{ij} d \log(\xi_j - \xi_i) + \text{regular}.\end{aligned}$$

Pairs not entirely contained in S remain regular in t . Summing over $i < j \in S$ gives the residue formula.

Theorem 7 (Transfer of flat spectral connections; Proved here) *Let*

$$(H, d_H) \xrightarrow{i} (V, d_V) \xrightarrow{p} (H, d_H)$$

be a filtered strong deformation retract with homotopy h :

$$pi = \text{id}_H, \quad ip = \text{id}_V + d_V h + h d_V.$$

Assume that $V \otimes \Omega^\bullet(\text{Conf}_n^{\text{ord}})$ carries a flat total differential

$$\mathbb{D}_V = (d_V + d_{\text{dR}}) - A_V,$$

where A_V lowers the filtration. Then the homological perturbation series converges and produces a flat total differential

$$\mathbb{D}_H = (d_H + d_{\text{dR}}) - A_H, \quad A_H = p A_V (1 - h A_V)^{-1} i = \sum_{m \geq 0} p A_V (h A_V)^m i.$$

Thus every flat bar-level spectral connection descends functorially to cohomology or to a minimal model.

Tensor the strong deformation retract with the de Rham complex of the base. Since d_{dR} commutes with i, p, h , the same data form a strong deformation retract for the base differential $d_V + d_{\text{dR}}$. Set $\delta = -A_V$. Because δ lowers the filtration, the standard homological perturbation series converges. The perturbation lemma gives the transferred differential

$$(d_H + d_{\text{dR}}) + p\delta(1 - h\delta)^{-1}i,$$

which squares to zero because $(d_V + d_{\text{dR}} + \delta)^2 = \mathbb{D}_V^2 = 0$. Replacing δ by $-A_V$ yields the formula for A_H .

1.3 Quasi-factorization and spectral obstruction theory

Definition 8 (Regularized holonomy data) *Assume that the flat connections of Theorem ?? are pronilpotent at the boundary. For $n = 2$, let*

$$R(u) \in 1 + F^1 A^{\widehat{\otimes} 2}$$

be the regularized holonomy from the positive tangential basepoint at collision to the interior point with difference $u = z_2 - z_1$. For $n = 3$, let $\Phi(w, z) \in 1 + F^2 A^{\widehat{\otimes}^3}$ be the regularized parallel transport between the two tangential basepoints corresponding to the bracketings (12)3 and 1(23) in $\overline{\text{Conf}}_3^{\text{ord}}(\mathbb{A}^1)$. We call (R, Φ) the spectral quasi-factorization datum.

Theorem 9 (Quasi-factorization theorem; Proved here) *Every residue-bounded complete dg-shifted Yangian equipped with a bar-horizontal strictification canonically determines a spectral quasi-factorization quantum group. More precisely, the regularized holonomy data (R, Φ) satisfy:*

1. quasi-hexagon relations on $\overline{\text{Conf}}_3^{\text{ord}}$, and
2. the pentagon relation on the codimension-one Stasheff cell complex of $\overline{\text{Conf}}_4^{\text{ord}}$.

The codimension-one boundary strata of $\overline{\text{Conf}}_3^{\text{ord}}$ corresponding to the bracketings (12)3 and 1(23) admit collar charts in which the regularized holonomy factors as the ordered product of the two-body holonomies. The comparison between these two collar factorizations is exactly the comparison holonomy $\Phi(w, z)$, which yields the quasi-hexagon.

For $n = 4$, the codimension-one boundary strata form the Stasheff pentagon. Flatness of the four-point connection implies that the total regularized transport around the boundary of that pentagon is the identity. Writing that transport as the product of the five comparison holonomies gives the pentagon relation.

Definition 10 (Spectral cochain complex and spectral Drinfeld class)
Let

$$C_{\text{spec}}^2(A) = \{g(u) \in A^{\widehat{\otimes}^2}\}, \quad C_{\text{spec}}^3(A) = \{\phi(w, z) \in A^{\widehat{\otimes}^3}\}.$$

Define the linearized spectral differential by

$$(\delta g)(w, z) = 1 \otimes g(z) + (\text{id} \otimes \Delta_z)g(w) - (\Delta_w \otimes \text{id})g(z) - g(w) \otimes 1.$$

The degree-three cohomology of the resulting complex is denoted by $H_{\text{spec}}^3(A)$. If

$$\Phi(w, z) = 1 + \phi_n(w, z) \pmod{F^{n+1}}$$

with ϕ_n the first nontrivial term, then its class

$$\mathfrak{D}_{\text{spec}}(A, \Omega) := [\phi_n] \in H_{\text{spec}}^3(\text{gr}^n A)$$

is called the spectral Drinfeld class.

Remark 11 (Well-definedness of the spectral Drinfeld class) *The spectral Drinfeld class is well-defined because the Hochschild-type obstruction cocycle it measures is independent of the choice of filtration splitting: different splittings produce cohomologous cocycles, so the class in $H_{\text{spec}}^3(\text{gr}^n A)$ is intrinsic.*

1.4. LOW-ORDER ALGEBRA: QUADRATIC AND CUBIC CONTROL 7

Theorem 12 (Strictification by spectral cohomology; Proved here) *Suppose*

$$\Phi(w, z) = 1 + \phi_n(w, z) \pmod{F^{n+1}}$$

with $n \geq 2$. Then:

1. ϕ_n is a 3-cocycle in the spectral complex;
2. under a spectral twist $G_n(u) = 1 + g_n(u) \pmod{F^{n+1}}$ one has

$$\phi_n \longmapsto \phi_n + \delta g_n;$$

3. if $[\phi_n] = 0$ in $H_{\text{spec}}^3(\text{gr}^n A)$, then a twist exists which kills ϕ_n modulo F^{n+1} ;
4. if $H_{\text{spec}}^3(\text{gr}^n A) = 0$ for all $n \geq 2$, then there is a filtered twist G for which $\Phi^G = 1$, hence the quasi-factorization datum strictifies to an honest spectral factorization R -matrix.

Expand the pentagon relation in the first nontrivial filtration degree. All nonlinear terms are of higher filtration, so the degree- n term satisfies the linearized pentagon, i.e. it is a cocycle. Expanding the standard twist formula for the quasi-associator to degree n yields the addition by the coboundary δg_n . Killing the first nontrivial term and iterating recursively proves the last two statements.

Remark 13 (First conceptual reduction) *The dg-shifted Yangian/factorization bridge is not a binary question. There is first a canonical passage from bar-horizontal strictification data for a dg-shifted Yangian to a spectral quasi-factorization package,*

dg-shifted Yangian + bar-horizontal strictification $\longrightarrow$ spectral quasi-factorization quantum group,

and only then a strictification problem. The latter is controlled by a concrete filtered cohomology theory.

1.4 Low-order algebra: quadratic and cubic control

Proposition 14 (Quadratic hexagon obstruction; Proved here) *Let $\tilde{r}(z)$ be the suspended degree-zero kernel. Define the naive multiplicative two-jet*

$$R^{(2)}(z) = 1 + \tilde{r}(z) + \frac{1}{2}\tilde{r}(z)^{\star 2}.$$

Assume the infinitesimal coassociativity identities

$$\begin{aligned} (\text{id} \otimes \Delta_z)\tilde{r}(z + w) &= \tilde{r}_{12}(w) + \tilde{r}_{13}(z + w), \\ (\Delta_w \otimes \text{id})\tilde{r}(z) &= \tilde{r}_{13}(z + w) + \tilde{r}_{23}(z). \end{aligned}$$

Then modulo F^3 ,

$$\begin{aligned} (\mathrm{id} \otimes \Delta_z) R^{(2)}(z+w) - R_{12}^{(2)}(w) R_{13}^{(2)}(z+w) &\equiv \frac{1}{2} [\tilde{r}_{13}(z+w), \tilde{r}_{12}(w)], \\ (\Delta_w \otimes \mathrm{id}) R^{(2)}(z) - R_{13}^{(2)}(z+w) R_{23}^{(2)}(z) &\equiv \frac{1}{2} [\tilde{r}_{23}(z), \tilde{r}_{13}(z+w)]. \end{aligned}$$

Expand both sides to second order and subtract. The square terms cancel except for the order-sensitive cross terms, which produce the displayed commutators.

Proposition 15 (Three-particle obstruction identity; Proved here) *Let $B^{\leq 3}(Y)$ be the reduced bar coalgebra truncated at tensor length 3, and let $I_{\tilde{r}}$ be the adjacent-gap insertion coderivation determined by $\tilde{r}$. Put*

$$D = d_B + I_{\tilde{r}}.$$

Assume the one-body and two-body Maurer–Cartan identities already hold. Then on tensor length 3,

$$D^2 = I_{\mathfrak{Y}_{\tilde{r}}},$$

where

$$\mathfrak{Y}_{\tilde{r}}(z, w) = MC(\tilde{r}_{12}(w) + \tilde{r}_{13}(z+w) + \tilde{r}_{23}(z)) - MC(\tilde{r}_{12}(w)) - MC(\tilde{r}_{13}(z+w)) - MC(\tilde{r}_{23}(z)).$$

Thus the tensor-length-3 obstruction is exactly the A_∞ Yang–Baxter defect.

Expand

$$D^2 = [d_B, I_{\tilde{r}}] + I_{\tilde{r}}^2.$$

On length 3, disjoint insertions cancel by coderivation signs. The surviving terms are exactly the overlapping insertions in the three ordered gaps. These assemble into the bar realization of the displayed Maurer–Cartan defect.

Theorem 16 (Abelian strictification theorem; Proved here) *Assume that all coefficients of $\tilde{r}(z)$ supercommute after suspension, that $m_k = 0$ for $k \geq 3$, and that the underlying coproduct is cocommutative. Then*

$$R(z) = \exp_\star(\tilde{r}(z))$$

converges and defines an honest spectral R -matrix on the completed bialgebra underlying Y .

The infinitesimal coassociativity identities exponentiate because all coefficients commute. The hexagon identities for the exponentiated coproduct follow from the BCH formula: since all coefficients commute, BCH reduces to ordinary addition, and the hexagon relation exponentiates termwise. Cocommutativity implies almost cocommutativity automatically in the commuting completion.

Theorem 17 (Residue-bounded completion theorem; Proved here) *Let $Y = \varprojlim_N Y_N$ be residue-bounded and complete, where $Y_N = Y/F^{N+1}Y$. Assume:*

1. *each Y_N is quadratic-Koszul;*
2. *the induced kernels satisfy the three-particle identity above;*
3. *the inverse system $H^m(B(Y_N))$ is Mittag-Leffler for every m .*

Then the completed coderivation

$$D = d_B + I_{\tilde{r}}$$

converges continuously on $\widehat{B}(Y)$, satisfies $D^2 = 0$, and the completed cobar comparison

$$\Omega\widehat{B}(Y) \longrightarrow Y$$

is a quasi-isomorphism.

Residue-boundedness implies that on each fixed filtration quotient only finitely many residues contribute, hence the coderivation converges. The three-particle obstruction identity implies $D^2 = 0$ quotientwise; separatedness of the filtration ($\bigcap_n F^n = 0$ by the pole-order bound) passes the identity to the inverse limit. Quadratic-Koszulness gives quotientwise bar-cobar quasi-isomorphisms (by [?, Proposition 2.2.1]), and the Mittag-Leffler hypothesis (which holds because the transition maps $H^m(B(Y_{N+1})) \rightarrow H^m(B(Y_N))$ are surjective) passes quasi-isomorphism to the inverse limit.

1.5 Exact Cartan transport and the first genuine shifted examples

[Cartan shift cocycle] Let H be a completed commutative primitive Cartan bialgebra with primitive generators $h_1, \dots, h_\ell$. Suppose the shifted Cartan currents are

$$H_i(u) = \frac{\prod_j (u - a_j^{(i)})}{\prod_k (u - b_k^{(i)})}.$$

Set

$$\phi_i(u) = \log H_i(u).$$

Define the formal bicharacter

$$F_\mu(u) = \exp\left(\sum_{i=1}^{\ell} \phi_i(u) h_i \otimes h_i\right).$$

Proposition 18 (Cartan shift cocycle identities; Proved here) *The formal bicharacter $F_\mu(u)$ satisfies*

$$\begin{aligned} (\text{id} \otimes \Delta)F_\mu(u) &= F_\mu^{12}(u)F_\mu^{13}(u), \\ (\Delta \otimes \text{id})F_\mu(u) &= F_\mu^{13}(u)F_\mu^{23}(u), \end{aligned}$$

and is additive in the shift parameter whenever the logarithms are.

Since $\Delta h_i = h_i \otimes 1 + 1 \otimes h_i$ and the h_i commute, the exponent splits exactly under either iterated coproduct.

Theorem 19 (Cartan gauge-twist theorem; Proved here) *Let*

$$\mathbb{D}_n^0 = (d + d_{\text{dR}}) - A_n^0$$

be an unshifted derived KZ differential. Suppose that on the chosen sector the operators $h_i^{(a)}h_i^{(b)}$ commute with A_n^0 . Set

$$S_{\mu,n} = \sum_{1 \leq a < b \leq n} \sum_{i=1}^{\ell} \phi_i(z_b - z_a) h_i^{(a)} h_i^{(b)}, \quad G_{\mu,n} = e^{-S_{\mu,n}}.$$

Then

$$\mathbb{D}_n^\mu = G_{\mu,n} \mathbb{D}_n^0 G_{\mu,n}^{-1} = (d + d_{\text{dR}}) - A_n^\mu$$

has connection form

$$A_n^\mu = A_n^0 + \sum_{1 \leq a < b \leq n} \sum_{i=1}^{\ell} \phi'_i(z_b - z_a) h_i^{(a)} h_i^{(b)} d(z_b - z_a).$$

The h_i commute in the dg-shifted setting because the Cartan subalgebra $\mathfrak{h} \subset \mathfrak{g}$ is abelian, and the dg-shifted coproduct preserves commutativity. Because the exponents commute on the sector in question,

$$dG_{\mu,n} G_{\mu,n}^{-1} = -dS_{\mu,n}.$$

The stated commutation hypothesis implies $G_{\mu,n} A_n^0 G_{\mu,n}^{-1} = A_n^0$, so the gauge transform contributes exactly $dS_{\mu,n}$.

Theorem 20 (Cartan-diagonal shift defect is exact; Proved here) *Fix a Cartan-diagonal line L_λ and write*

$$\phi_i(u) = \log H_i(u).$$

The shifted two-body scalar factor is

$$R_{\mu,\lambda}^{\text{sh}}(u) = \exp\left(\sum_i \lambda_i^2 \phi_i(u)\right).$$

The three-body quasi-hexagon defect is therefore

$$\Phi_{\mu,\lambda}^{\text{sh}}(w, z) = \frac{R_{\mu,\lambda}^{\text{sh}}(w + z)}{R_{\mu,\lambda}^{\text{sh}}(w)R_{\mu,\lambda}^{\text{sh}}(z)},$$

and its logarithm is

$$\mathfrak{D}_{\text{spec},\mu}^{\lambda}(w, z) = \sum_i \lambda_i^2 (\phi_i(w + z) - \phi_i(w) - \phi_i(z)).$$

Hence the Cartan-diagonal contribution to the spectral Drinfeld class is exact.

Immediate from the definition of the shifted scalar transport.

Remark 21 (First genuine shifted examples) *The framed A_1 shifted Yangian and the staircase shifted Yangians of finite type A furnish the first genuine shifted examples. In all such Cartan-diagonal sectors, the shift produces a visible three-body logarithmic defect but no cohomological obstruction class.*

1.6 Filtration 2: adjacent-root triangles

The first genuinely mixed-root sector arises from adjacent simple roots. Let $i \sim j$ be adjacent simple roots in a rank-2 Levi of type A_2 . Set

$$E_{ij} = [E_i, E_j], \quad F_{ij} = [F_j, F_i],$$

so that $[F_i, F_j] = -F_{ij}$. For $a \neq b$, write

$$X_i^{ab}(u) = \rho_i(u) E_i^{(a)} F_i^{(b)}, \quad X_{ij}^{ab}(u) = \rho_{ij}(u) E_{ij}^{(a)} F_{ij}^{(b)}.$$

Definition 22 (Triangle sectors) *On three tensor factors define the left and right adjacent-root triangle vectors*

$$T_{ij}^L = E_{ij}^{(1)} F_i^{(2)} F_j^{(3)}, \quad T_{ij}^R = E_i^{(1)} E_j^{(2)} F_{ij}^{(3)}.$$

The corresponding opposite orientations are

$$T_{ji}^L = E_{ij}^{(1)} F_j^{(2)} F_i^{(3)}, \quad T_{ji}^R = E_j^{(1)} E_i^{(2)} F_{ij}^{(3)}.$$

Theorem 23 (Triangle localization formula; Proved here) *Let*

$$S_{12}(w) = \rho_i(w) E_i^{(1)} F_i^{(2)} + \rho_j(w) E_j^{(1)} F_j^{(2)} + \cdots,$$

$$S_{13}(w + z) = \rho_i(w + z) E_i^{(1)} F_i^{(3)} + \rho_j(w + z) E_j^{(1)} F_j^{(3)} + \cdots.$$

Then

$$\Pi_{\langle T_{ij}^L, T_{ji}^L \rangle} \frac{1}{2} [S_{12}(w), S_{13}(w + z)] = \frac{1}{2} \rho_i(w) \rho_j(w + z) T_{ij}^L - \frac{1}{2} \rho_j(w) \rho_i(w + z) T_{ji}^L.$$

Similarly,

$$\Pi_{\langle T_{ij}^R, T_{ji}^R \rangle} \frac{1}{2} [S_{13}(w + z), S_{23}(z)] = -\frac{1}{2} \rho_i(w + z) \rho_j(z) T_{ij}^R + \frac{1}{2} \rho_j(w + z) \rho_i(z) T_{ji}^R.$$

The only contributing commutators are

$$[E_i^{(1)} F_i^{(2)}, E_j^{(1)} F_j^{(3)}] = E_{ij}^{(1)} F_i^{(2)} F_j^{(3)},$$

and

$$[E_j^{(1)} F_j^{(2)}, E_i^{(1)} F_i^{(3)}] = -E_{ij}^{(1)} F_j^{(2)} F_i^{(3)},$$

with the analogous computation on the right using $[F_i, F_j] = -F_{ij}$.

Definition 24 (Triangle coproduct coefficients) *Let Δ_z be the twisted coproduct. Define the triangle transport coefficients by*

$$\Pi_{(T_{ij}^L, T_{ji}^L)}((\text{id} \otimes \Delta_z) X_{ij}^{12}(w)) = \kappa_{ij}^L(w, z) T_{ij}^L - \kappa_{ji}^L(w, z) T_{ji}^L,$$

and similarly on the right,

$$\Pi_{(T_{ij}^R, T_{ji}^R)}((\Delta_w \otimes \text{id}) X_{ij}^{23}(z)) = -\kappa_{ij}^R(w, z) T_{ij}^R + \kappa_{ji}^R(w, z) T_{ji}^R.$$

Theorem 25 (Adjacent-root rigidity; Proved here) *The filtration-2 hexagon defect vanishes in the oriented triangle sectors if and only if*

$$\begin{aligned} \rho_{ij}(w) \kappa_{ij}^L(w, z) &= \frac{1}{2} \rho_i(w) \rho_j(w + z), \\ \rho_{ij}(w) \kappa_{ji}^L(w, z) &= \frac{1}{2} \rho_j(w) \rho_i(w + z), \\ \rho_{ij}(z) \kappa_{ij}^R(w, z) &= \frac{1}{2} \rho_i(w + z) \rho_j(z), \\ \rho_{ij}(z) \kappa_{ji}^R(w, z) &= \frac{1}{2} \rho_j(w + z) \rho_i(z). \end{aligned}$$

Project the logarithm of the product side to the triangle sectors. The only contributions are the commutator terms computed in Theorem ?? . On the coproduct side, X_{ij} is the unique root generator whose twisted coproduct can land in those sectors. Equality of coefficients gives the claim.

Corollary 26 (Normalized rational triangle coefficient; Proved here)
Assume the rational ansatz

$$\rho_i(u) = \frac{c_i}{u}, \quad \rho_j(u) = \frac{c_j}{u}, \quad \rho_{ij}(u) = \frac{c_{ij}}{u},$$

and

$$\kappa_{ij}^L(w, z) = \frac{\lambda_{ij}^L}{w + z}, \quad \kappa_{ij}^R(w, z) = \frac{\lambda_{ij}^R}{w + z},$$

with the same form for the opposite orientations. Then

$$\lambda_{ij}^L = \lambda_{ji}^L = \lambda_{ij}^R = \lambda_{ji}^R = \frac{c_i c_j}{2c_{ij}}.$$

In the normalized case $c_i = c_j = c_{ij} = 1$, the universal coefficient is $\frac{1}{2}$.

Substitute the rational ansatz into the equations of Theorem ??.

Remark 27 (Fortification: transport, not naive invariance) *A naive statement of “shift-invariance” at mixed-root level is too coarse and, in general, false if interpreted coefficientwise. The shift changes the local mixed-root coefficients by explicit ratios of shift characters at different spectral arguments. What is invariant is not the raw coefficient but the obstruction class after transport by the shift cocycle.*

Proposition 28 (Triangle shift transport law; Proved here) *Suppose the shifted root coefficients satisfy*

$$\rho_\alpha^\mu(u) = \chi_\alpha^\mu(u) \rho_\alpha^0(u)$$

for scalar shift characters $\chi_\alpha^\mu(u)$. Assume the unshifted triangle coefficients solve the rigidity equations. Then the shifted coefficients are uniquely forced to be

$$\begin{aligned} \kappa_{ij}^{L,\mu}(w, z) &= \frac{\chi_i^\mu(w) \chi_j^\mu(w+z)}{\chi_{ij}^\mu(w)} \kappa_{ij}^{L,0}(w, z), \\ \kappa_{ij}^{R,\mu}(w, z) &= \frac{\chi_i^\mu(w+z) \chi_j^\mu(z)}{\chi_{ij}^\mu(z)} \kappa_{ij}^{R,0}(w, z), \end{aligned}$$

with the analogous formulas for the opposite orientations.

Insert the shifted coefficients into the identities of Theorem ?? and solve for $\kappa^{L,\mu}$ and $\kappa^{R,\mu}$.

Interpretation 29 *The first mixed-root sector already detects shift sensitivity at the level of local transport functions. But this sensitivity is completely forced by the shifted one-body characters; it does not define a new independent obstruction class.*

1.7 The exact free two-body model

The free example of [?] admits a particularly transparent computation. Write

$$X(x) = \sum_{n \geq 0} X_n x^n, \quad \Psi(x) = \sum_{n \geq 0} \psi_n x^n.$$

The dg kernel in the free model is

$$r(u) = \sum_{n,m \geq 0} (-1)^n \binom{n+m}{m} \frac{\psi_n \otimes X_m - X_n \otimes \psi_m}{u^{n+m+1}}.$$

Lemma 30 (Closed generating kernel; Proved here) *Coefficientwise in x, y ,*

$$\sum_{n,m \geq 0} (-1)^n \binom{n+m}{m} \frac{x^n y^m}{u^{n+m+1}} = \frac{1}{u+x-y}.$$

Hence the free suspended kernel is the coefficientwise expansion of

$$\mathcal{R}_{\text{free}}(u; x, y) = \frac{\Psi(x) \otimes X(y) - X(x) \otimes \Psi(y)}{u+x-y}.$$

Summing over $N = n + m$ gives

$$\begin{aligned} \sum_{n,m \geq 0} (-1)^n \binom{n+m}{m} \frac{x^n y^m}{u^{n+m+1}} &= \frac{1}{u} \sum_{N \geq 0} \frac{1}{u^N} \sum_{m=0}^N \binom{N}{m} (-x)^{N-m} y^m \\ &= \frac{1}{u} \sum_{N \geq 0} \left(\frac{y-x}{u} \right)^N = \frac{1}{u+x-y}. \end{aligned}$$

Theorem 31 (Exact free two-body transport; Proved here) *Let*

$$\Theta(x, y) = \Psi(x) \otimes X(y) - X(x) \otimes \Psi(y).$$

In the free supercommutative model, $\Theta(x, y)^2 = 0$. Therefore the connection

$$\nabla_{\text{free}} = d - \frac{\Theta(x, y)}{u+x-y} du$$

integrates exactly with transport

$$T(u, u_0) = 1 + \Theta(x, y) \log \frac{u+x-y}{u_0+x-y}.$$

The local monodromy around the formal divisor $u+x-y=0$ is

$$M_{\text{loop}} = 1 + 2\pi i \Theta(x, y).$$

Because $\Theta^2 = 0$,

$$\exp \left(\Theta \log \frac{u+x-y}{u_0+x-y} \right) = 1 + \Theta \log \frac{u+x-y}{u_0+x-y}.$$

Differentiating gives the transport equation. The monodromy follows by adding $2\pi i$ to the logarithm around a loop.

1.8 Filtration 3: root quadrilaterals

We now perform the first genuinely class-3 computation. Let $i \sim j \sim k$ be a chain of simple roots of type A_3 with i not adjacent to k . Define the composite roots by

$$E_{ij} = [E_i, E_j], \quad E_{jk} = [E_j, E_k], \quad E_{ijk} = [E_{ij}, E_k] = [E_i, E_{jk}],$$

and for the opposite side set

$$F_{ij} = [F_j, F_i], \quad F_{jk} = [F_k, F_j], \quad F_{ijk} = [F_k, F_{ij}] = [F_{jk}, F_i].$$

Then

$$[F_i, F_j] = -F_{ij}, \quad [F_j, F_k] = -F_{jk}, \quad [F_i, F_{jk}] = -[F_k, F_{ij}] = -F_{ijk}.$$

Definition 32 (Root-quadrilateral sectors) *On four tensor factors define the oriented left and right quadrilateral vectors*

$$Q_{ijk}^L = E_{ijk}^{(1)} F_i^{(2)} F_j^{(3)} F_k^{(4)}, \quad Q_{ijk}^R = E_i^{(1)} E_j^{(2)} E_k^{(3)} F_{ijk}^{(4)}.$$

The opposite orientation is obtained by replacing (i, j, k) by (k, j, i) and the sign conventions above.

Lemma 33 (Class-3 ordered BCH coefficient; Proved here) *Let X, Y, Z lie in a class-3 nilpotent Lie algebra, so all commutators of depth 4 vanish. The multilinear XYZ -part of $\log(e^X e^Y e^Z)$ is*

$$\frac{1}{4}[[X, Y], Z] + \frac{1}{12}[X, [Y, Z]] + \frac{1}{12}[Y, [X, Z]].$$

If, in addition, $[X, Z] = 0$, then this simplifies to

$$\text{mult}_{XYZ} \log(e^X e^Y e^Z) = \frac{1}{3}[X, [Y, Z]].$$

Write

$$U = \log(e^X e^Y) = X + Y + \frac{1}{2}[X, Y] + \frac{1}{12}[X, [X, Y]] + \frac{1}{12}[Y, [Y, X]].$$

Now compute

$$\log(e^X e^Y e^Z) = \log(e^U e^Z) = U + Z + \frac{1}{2}[U, Z] + \frac{1}{12}[U, [U, Z]] + \frac{1}{12}[Z, [Z, U]].$$

Retaining only the multilinear XYZ -part gives the first displayed formula. If $[X, Z] = 0$, then

$$[[X, Y], Z] = [X, [Y, Z]]$$

by Jacobi, and the second formula follows.

Theorem 34 (Quadrilateral localization formula; Proved here) *Let*

$$X_i^{12}(u) = \rho_i(u) E_i^{(1)} F_i^{(2)}, \quad X_j^{13}(u+v) = \rho_j(u+v) E_j^{(1)} F_j^{(3)}, \quad X_k^{14}(u+v+t) = \rho_k(u+v+t) E_k^{(1)} F_k^{(4)}.$$

Then the projection of

$$\log \left(e^{X_i^{12}(u)} e^{X_j^{13}(u+v)} e^{X_k^{14}(u+v+t)} \right)$$

to the left quadrilateral sector is

$$\frac{1}{3} \rho_i(u) \rho_j(u+v) \rho_k(u+v+t) Q_{ijk}^L.$$

Similarly, the projection of

$$\log \left(e^{\rho_i(u+v+t) E_i^{(1)} F_i^{(4)}} e^{\rho_j(v+t) E_j^{(2)} F_j^{(4)}} e^{\rho_k(t) E_k^{(3)} F_k^{(4)}} \right)$$

to the right quadrilateral sector is

$$-\frac{1}{3} \rho_i(u+v+t) \rho_j(v+t) \rho_k(t) Q_{ijk}^R.$$

For the left sector, set

$$X = X_i^{12}(u), \quad Y = X_j^{13}(u+v), \quad Z = X_k^{14}(u+v+t).$$

Since i and k are nonadjacent, $[X, Z] = 0$. By Lemma ??, the multilinear part is $\frac{1}{3}[X, [Y, Z]]$. Now

$$[Y, Z] = \rho_j(u+v) \rho_k(u+v+t) E_{jk}^{(1)} F_j^{(3)} F_k^{(4)},$$

and then

$$[X, [Y, Z]] = \rho_i(u) \rho_j(u+v) \rho_k(u+v+t) E_{ijk}^{(1)} F_i^{(2)} F_j^{(3)} F_k^{(4)}.$$

This proves the first formula.

For the right sector, set

$$X = \rho_i(u+v+t) E_i^{(1)} F_i^{(4)}, \quad Y = \rho_j(v+t) E_j^{(2)} F_j^{(4)}, \quad Z = \rho_k(t) E_k^{(3)} F_k^{(4)}.$$

Again $[X, Z] = 0$, so the multilinear part is $\frac{1}{3}[X, [Y, Z]]$. Here

$$[Y, Z] = -\rho_j(v+t) \rho_k(t) E_j^{(2)} E_k^{(3)} F_{jk}^{(4)},$$

so

$$[X, [Y, Z]] = -\rho_i(u+v+t) \rho_j(v+t) \rho_k(t) E_i^{(1)} E_j^{(2)} E_k^{(3)} F_{ijk}^{(4)}.$$

This is the second formula.

Definition 35 (Quadrilateral coproduct coefficients) Let $\Delta_{v,t}^{(2)}$ denote the two-step twisted coproduct separating one factor into three ordered factors with spectral gaps v, t . Define the left and right quadrilateral transport coefficients by

$$\Pi_{\langle Q_{ijk}^L \rangle}((\text{id} \otimes \Delta_{v,t}^{(2)})X_{ijk}^{12}(u)) = \Lambda_{ijk}^L(u, v, t)Q_{ijk}^L,$$

$$\Pi_{\langle Q_{ijk}^R \rangle}((\Delta_{u,v}^{(2)} \otimes \text{id})X_{ijk}^{34}(t)) = \Lambda_{ijk}^R(u, v, t)Q_{ijk}^R.$$

Theorem 36 (Quadrilateral rigidity; Proved here) The filtration-3 defect vanishes in the oriented quadrilateral sectors if and only if

$$\begin{aligned} \rho_{ijk}(u)\Lambda_{ijk}^L(u, v, t) &= \frac{1}{3}\rho_i(u)\rho_j(u+v)\rho_k(u+v+t), \\ \rho_{ijk}(t)\Lambda_{ijk}^R(u, v, t) &= \frac{1}{3}\rho_i(u+v+t)\rho_j(v+t)\rho_k(t). \end{aligned}$$

Project the logarithm of the ordered product of the relevant two-body factors to the quadrilateral sector. By Theorem ??, the product side has the stated coefficient. On the coproduct side, a quadrilateral vector with three distinct simple lowering or raising operators in separate tensor factors can only arise from the triple-root generator X_{ijk} : any simple-root or two-root composite generator remains supported in at most two root colors and cannot land in Q_{ijk}^L or Q_{ijk}^R after projection. Equality of coefficients yields the claim.

Corollary 37 (Normalized rational quadrilateral coefficient; Proved here)
Assume the rational ansatz

$$\rho_i(u) = \frac{c_i}{u}, \quad \rho_j(u) = \frac{c_j}{u}, \quad \rho_k(u) = \frac{c_k}{u}, \quad \rho_{ijk}(u) = \frac{c_{ijk}}{u}.$$

Suppose further that the quadrilateral coefficients have translation-covariant form

$$\Lambda_{ijk}^L(u, v, t) = \frac{\lambda_{ijk}^L}{(u+v)(u+v+t)}, \quad \Lambda_{ijk}^R(u, v, t) = \frac{\lambda_{ijk}^R}{(v+t)(u+v+t)}.$$

Then

$$\lambda_{ijk}^L = \lambda_{ijk}^R = \frac{c_i c_j c_k}{3c_{ijk}}.$$

In the normalized case $c_i = c_j = c_k = c_{ijk} = 1$, the universal quadrilateral coefficient is

$$\boxed{\frac{1}{3}}.$$

Substitute the rational ansatz into the identities of Theorem ??.

Proposition 38 (Quadrilateral shift transport law; Proved here) *Assume again that the shifted one-body coefficients factor as*

$$\rho_\alpha^\mu(u) = \chi_\alpha^\mu(u) \rho_\alpha^0(u).$$

If the unshifted coefficients solve Theorem ??, then the shifted quadrilateral transport coefficients are uniquely forced to be

$$\begin{aligned}\Lambda_{ijk}^{L,\mu}(u, v, t) &= \frac{\chi_i^\mu(u) \chi_j^\mu(u+v) \chi_k^\mu(u+v+t)}{\chi_{ijk}^\mu(u)} \Lambda_{ijk}^{L,0}(u, v, t), \\ \Lambda_{ijk}^{R,\mu}(u, v, t) &= \frac{\chi_i^\mu(u+v+t) \chi_j^\mu(v+t) \chi_k^\mu(t)}{\chi_{ijk}^\mu(t)} \Lambda_{ijk}^{R,0}(u, v, t).\end{aligned}$$

Insert the shifted coefficients into the two equations of Theorem ?? and solve for $\Lambda^{L,\mu}$ and $\Lambda^{R,\mu}$.

Corollary 39 (No new independent shift obstruction in filtration 3; Proved here)

A new shift-sensitive function first appears in filtration 3 through the transport laws of Proposition ??; however, this function is entirely forced by the shifted one-body characters. Thus no new independent obstruction class appears in the oriented root-quadrilateral sector.

The quadrilateral coefficient is uniquely determined by lower data once Theorem ?? holds. Therefore its shift variation contributes no new free parameter and no new independent cohomology class.

Remark 40 (First place the frontier moves) *The corrected picture is now sharp.*

1. Cartan-diagonal sectors produce exact defects.
2. Commuting sectors strictify directly.
3. Adjacent-root triangles are forced by a universal coefficient $\frac{1}{2}$, transported explicitly by shift characters.
4. Root quadrilaterals are forced by a universal coefficient $\frac{1}{3}$, again transported explicitly by shift characters.

At the level of this low-order analysis, the first place where a genuinely new and independent obstruction could live is in filtration ≥ 4 , where new Hall words and higher Magnus/BCH terms enter: the D_4 star, for instance, gives a two-dimensional free multilinear Lie space. The resolution comes in §??–§??: root multiplicity one — the defining representation-theoretic property of simple Lie algebras — collapses every root sector to one dimension, closing this frontier universally. The low-order coefficients $\frac{1}{2}$ and $\frac{1}{3}$ are the first instances of the universal $\frac{1}{n}$, but the vanishing of the obstruction does not depend on these numerical values.

1.9 Conditional global strictification criterion

Theorem 41 (Conditional strictification criterion; Proved here) *Assume:*

1. *the unshifted dg object strictifies to a factorization quantum group with spectral R-matrix R_0 ;*
2. *the shift is implemented by the Cartan cocycle of Proposition ??;*
3. *the infinitesimal logarithm of the shifted spectral transport agrees with the dg kernel after residue-bounded completion; and*
4. *the spectral Drinfeld classes vanish.*

Then the shifted dg-shifted Yangian admits a completed E_1 -chiral/factorization realization whose representation category is that of an honest spectral factorization quantum group.

The unshifted strictification gives a strict spectral object. Transport by the Cartan cocycle gives the shifted quasi-factorization transport. Vanishing of the spectral Drinfeld class then strictifies the shifted quasi-object. Residue-bounded completion upgrades finite-stage compatibility to the full completed statement.

1.10 The strictification theorem

Path-type root sectors

Definition 42 (Path-type sector at filtration n) *Let $\mathfrak{g}$ be a semisimple Lie algebra. A path-type sector at filtration n is determined by a chain of n simple roots $\alpha_1, \alpha_2, \dots, \alpha_n$ in the Dynkin diagram such that α_i is adjacent to α_{i+1} for $1 \leq i \leq n-1$ and α_i is not adjacent to α_j for $|i-j| \geq 2$. (Such a chain exists whenever $\mathfrak{g}$ contains a type- A_n subdiagram.) Define the composite root generators by right-normed iterated brackets:*

$$E_{\alpha_1 \dots \alpha_k} := [E_{\alpha_1}, [E_{\alpha_2}, \dots, [E_{\alpha_{k-1}}, E_{\alpha_k}] \dots]] \quad (2 \leq k \leq n),$$

and similarly for the opposite generators $F_{\alpha_1 \dots \alpha_k}$. The left n -gon vector and right n -gon vector on $n+1$ tensor factors are

$$P_{\alpha_1 \dots \alpha_n}^L = E_{\alpha_1 \dots \alpha_n}^{(1)} F_{\alpha_1}^{(2)} F_{\alpha_2}^{(3)} \dots F_{\alpha_n}^{(n+1)}, \quad P_{\alpha_1 \dots \alpha_n}^R = E_{\alpha_1}^{(1)} E_{\alpha_2}^{(2)} \dots E_{\alpha_n}^{(n)} F_{\alpha_1 \dots \alpha_n}^{(n+1)}.$$

Remark 43 *At $n=2$ these recover the triangle vectors $T_{ij}^{L/R}$ of the adjacent-root computation. At $n=3$ they recover the quadrilateral vectors $Q_{ijk}^{L/R}$ (relabelling $(i, j, k) \mapsto (\alpha_1, \alpha_2, \alpha_3)$). The general case is the n -gon sector.*

The path one-dimensionality theorem

Theorem 44 (Path Lie algebra one-dimensionality; Proved here) *Let $L_{\text{path}}(n)$ be the Lie algebra with generators $X_1, \dots, X_n$ and relations $[X_i, X_j] = 0$ for $|i - j| \geq 2$. Then the multilinear degree- n component*

$$L_{\text{path}}^n(X_1, \dots, X_n) := \text{span}\{\text{Lie monomials involving each } X_i \text{ exactly once}\}$$

is one-dimensional, generated by the right-normed bracket

$$[X_1, [X_2, \dots, [X_{n-1}, X_n] \cdots]].$$

By induction on n .

Base case ($n = 2$). The multilinear degree-2 space is spanned by $[X_1, X_2]$.

Inductive step. Let M be a multilinear degree- n Lie monomial in $X_1, \dots, X_n$. Write $M = [L, R]$ where L involves variables indexed by $S \subset \{1, \dots, n\}$ and R involves $T = \{1, \dots, n\} \setminus S$.

Step 1: Connectivity. For $[L, R] \neq 0$, both S and T must be single connected subpaths of $1-2-\dots-n$. We prove that the Lie algebra generated by any disconnected subset of the path generators is a direct sum, hence its multilinear Lie monomials vanish.

Suppose $S \subseteq \{1, \dots, n\}$ decomposes as $S = S_1 \sqcup S_2$ with $\max(S_1) < \min(S_2)$ and $\max(S_1) + 1 < \min(S_2)$ (a gap in the path). Then $[X_a, X_c] = 0$ for all $a \in S_1, c \in S_2$ (since $|a - c| \geq 2$). The Lie algebra generated by $\{X_s : s \in S\}$ is therefore the direct sum $\text{Lie}(\{X_s : s \in S_1\}) \oplus \text{Lie}(\{X_s : s \in S_2\})$. A multilinear Lie monomial involving generators from both summands is zero (every bracket tree must eventually pair elements from different summands, producing a commutator that vanishes in a direct sum). Hence $L = 0$ whenever S is disconnected. The same argument applies to T and R .

Step 2: Interval partition. From Step 1, the only non-vanishing decompositions are $S = \{1, \dots, m\}, T = \{m+1, \dots, n\}$ for some $1 \leq m \leq n-1$ (or the reverse, but $[L, R] = -[R, L]$). By the induction hypothesis, L is proportional to $[X_1, [X_2, \dots, X_m] \cdots]$ and R is proportional to $[X_{m+1}, [X_{m+2}, \dots, X_n] \cdots]$. So M is proportional to

$$[[X_1, \dots, X_m], [X_{m+1}, \dots, X_n]]$$

(right-normed brackets on each side).

Step 3: Reduction to the standard generator. We show that

$$[[X_1, \dots, X_m], [X_{m+1}, \dots, X_n]] = [X_1, [X_2, \dots, [X_{n-1}, X_n] \cdots]]$$

in $L_{\text{path}}(n)$ for every $1 \leq m \leq n-1$. Apply the Jacobi identity:

$$\begin{aligned} [[X_1, \dots, X_m], [X_{m+1}, \dots, X_n]] &= [X_1, [[X_2, \dots, X_m], [X_{m+1}, \dots, X_n]]] \\ &\quad - [[X_2, \dots, X_m], [X_1, [X_{m+1}, \dots, X_n]]]. \end{aligned}$$

For the second term: $[X_1, X_j] = 0$ for all $j \geq 3$, hence $[X_1, [X_{m+1}, \dots, X_n]] = 0$ whenever $m \geq 2$ (the outermost bracket involves X_1 and a monomial on $\{X_{m+1}, \dots, X_n\}$; by induction on bracketing depth, every leaf of that monomial has $[X_1, X_j] = 0$ for $j \geq m+1 \geq 3$).

Thus the second term vanishes for $m \geq 2$, and we have

$$[[X_1, \dots, X_m], [X_{m+1}, \dots, X_n]] = [X_1, [[X_2, \dots, X_m], [X_{m+1}, \dots, X_n]]].$$

By the induction hypothesis applied to the subpath $\{2, \dots, n\}$ (which has $n-1$ generators satisfying the path relations), the inner bracket $[[X_2, \dots, X_m], [X_{m+1}, \dots, X_n]]$ is proportional to $[X_2, [X_3, \dots, X_n] \cdots]$. Hence

$$M \propto [X_1, [X_2, \dots, [X_{n-1}, X_n] \cdots]].$$

This is a single generator, so $\dim L_{\text{path}}^n = 1$.

The universal BCH coefficient via the Dynkin projection

Theorem 45 (Dynkin coefficient for the ordered path BCH; Proved here)

Let $X_1, \dots, X_n$ satisfy the path commutation relations $[X_i, X_j] = 0$ for $|i - j| \geq 2$. Then the multilinear $X_1 \cdots X_n$ -part of the ordered Baker–Campbell–Hausdorff product is

$$\text{mult}_{X_1 \cdots X_n} \log(e^{X_1} e^{X_2} \cdots e^{X_n}) = \frac{1}{n} [X_1, [X_2, \dots, [X_{n-1}, X_n] \cdots]]. \quad (1.1)$$

By Theorem ??, the multilinear part is $c_n \cdot [X_1, [X_2, \dots, X_n] \cdots]$ for a unique scalar c_n . We determine c_n by passing to the universal enveloping algebra and extracting the coefficient of the concatenation monomial $X_1 \cdot X_2 \cdots X_n$.

Step 1: Coefficient in the Lie bracket. In the universal enveloping algebra, the right-normed bracket $[X_1, [X_2, \dots, [X_{n-1}, X_n] \cdots]]$ expands as a signed sum of permuted monomials. Its leading term (in the order $X_1 X_2 \cdots X_n$) has coefficient $+1$.

Step 2: Coefficient in the BCH logarithm. In the free associative algebra, $e^{X_1} \cdot e^{X_2} \cdots e^{X_n}$ has multilinear part exactly $X_1 \cdot X_2 \cdots X_n$ (each X_i appears in a single exponential factor, so the multilinear extraction picks X_i from the degree-1 term of e^{X_i} and 1 from all other factors).

The multilinear part of $\log(P)$ where $P = e^{X_1} \cdots e^{X_n}$ is computed via $\log(1 + A) = \sum_{k \geq 1} (-1)^{k+1} A^k / k$ with $A = P - 1$. The subset-multilinear decomposition of A gives: the S -multilinear part of A (for $S \subseteq \{1, \dots, n\}$, $S \neq \emptyset$) is the ordered product $\prod_{i \in S}^{\rightarrow} X_i$. Therefore

$$\text{coeff of } X_1 \cdots X_n \text{ in } \log(P) = \sum_{k=1}^n \frac{(-1)^{k+1}}{k} \cdot \#\{\text{compositions of } n \text{ into } k \text{ parts}\},$$

because the monomial $X_1 \cdots X_n$ arises from A^k only when the k factors draw consecutive blocks $\{1, \dots, i_1\}$, $\{i_1+1, \dots, i_1+i_2\}$, etc. (all other set-partitions reorder the variables). The number of compositions of n into k parts is $\binom{n-1}{k-1}$.

Step 3: The integral identity. We evaluate the resulting sum using the beta integral:

$$\begin{aligned} \sum_{k=1}^n \frac{(-1)^{k+1}}{k} \binom{n-1}{k-1} &= \sum_{j=0}^{n-1} \frac{(-1)^j}{j+1} \binom{n-1}{j} \\ &= \sum_{j=0}^{n-1} \binom{n-1}{j} (-1)^j \int_0^1 t^j dt \\ &= \int_0^1 \sum_{j=0}^{n-1} \binom{n-1}{j} (-t)^j dt \\ &= \int_0^1 (1-t)^{n-1} dt = \frac{1}{n}. \end{aligned}$$

Step 4: Conclusion. Combining Steps 1–3: the coefficient of $X_1 \cdots X_n$ in $c_n \cdot [X_1, \dots, X_n]$ is $c_n \cdot 1$, and in $\log(P)$ is $\frac{1}{n}$. Hence $c_n = \frac{1}{n}$.

Remark 46 At $n = 2$ this gives $\frac{1}{2}[X_1, X_2]$; at $n = 3$, the coefficient $\frac{1}{3}[X_1, [X_2, X_3]]$ of Lemma ?? (in the path case $[X_1, X_3] = 0$). The integral identity $\int_0^1 (1-t)^{n-1} dt = 1/n$ replaces the case-by-case BCH algebra with a single uniform argument.

The all-filtration localization formula

Theorem 47 (n -gon localization formula; Proved here) For a path $\alpha_1 \sim \alpha_2 \sim \cdots \sim \alpha_n$ in the Dynkin diagram, set

$$X_i^{1, i+1}(u_i) = \rho_{\alpha_i}(u_i) E_{\alpha_i}^{(1)} F_{\alpha_i}^{(i+1)}, \quad u_i = u + \sum_{j=1}^{i-1} v_j,$$

where $(u, v_1, \dots, v_{n-1})$ are spectral parameters. Then the projection of

$$\log \left(e^{X_1^{1,2}(u_1)} e^{X_2^{1,3}(u_2)} \cdots e^{X_n^{1,n+1}(u_n)} \right)$$

to the left n -gon sector is

$$\frac{1}{n} \rho_{\alpha_1}(u_1) \rho_{\alpha_2}(u_2) \cdots \rho_{\alpha_n}(u_n) P_{\alpha_1 \cdots \alpha_n}^L. \quad (1.2)$$

The right-sector computation is dual: the lowering operators are gathered in one tensor factor (the $(n+1)$ -st), and the bracket convention $F_{\alpha\beta} = [F_\beta, F_\alpha]$ introduces a single overall minus sign. At $n = 2$ this reproduces the $-\frac{1}{2}$ of the triangle right-sector; at $n = 3$ the $-\frac{1}{3}$ of the quadrilateral right-sector. In general the right n -gon coefficient is $-1/n$.

By Theorem ??, the multilinear part of the ordered exponential is $\frac{1}{n}[X_1^{1,2}, [X_2^{1,3}, \dots, [X_{n-1}^{1,n}, X_n^{1,n+1}]] \cdots$

Left sector. The innermost bracket $[X_{n-1}^{1,n}, X_n^{1,n+1}]$ produces $E_{\alpha_{n-1}\alpha_n}^{(1)}$ in the first tensor factor (since $[E_{\alpha_{n-1}}, E_{\alpha_n}] = E_{\alpha_{n-1}\alpha_n}$) and the product $F_{\alpha_{n-1}}^{(n)} F_{\alpha_n}^{(n+1)}$

across the remaining factors. Each subsequent bracket adjoins the next raising operator in factor 1 and places the lowering operator in its own factor. At the top of the nesting, the first factor carries $E_{\alpha_1 \dots \alpha_n}$ and factor $(i+1)$ carries F_{α_i} , giving exactly $P_{\alpha_1 \dots \alpha_n}^L$. The coefficient is $\prod_{i=1}^n \rho_{\alpha_i}(u_i)$.

Right sector. The right n -gon vector P^R has all lowering operators in a single factor. The bracket convention $F_{\alpha\beta} = [F_\beta, F_\alpha]$ (the opposite nesting to $E_{\alpha\beta} = [E_\alpha, E_\beta]$) means that the nested commutator of F -generators in one tensor factor produces $(-1) \cdot F_{\alpha_1 \dots \alpha_n}$, giving an overall coefficient $-1/n$ times the one-body products.

n -gon rigidity and the all-filtration vanishing theorem

Definition 48 (n -gon coproduct coefficients) Define the left and right n -gon transport coefficients by

$$\Pi_{\langle PL \rangle} \left((\text{id} \otimes \Delta^{(n-1)}) X_{\alpha_1 \dots \alpha_n}^{1, \bullet}(u) \right) = \Lambda_{\alpha_1 \dots \alpha_n}^L(u, v_1, \dots, v_{n-1}) P_{\alpha_1 \dots \alpha_n}^L,$$

and analogously for Λ^R .

Theorem 49 (n -gon rigidity; Proved here) The filtration- $(n-1)$ defect vanishes in the oriented path-type n -gon sectors if and only if

$$\rho_{\alpha_1 \dots \alpha_n}(u) \Lambda_{\alpha_1 \dots \alpha_n}^L(u, v_1, \dots, v_{n-1}) = \frac{1}{n} \prod_{i=1}^n \rho_{\alpha_i}(u_i). \quad (1.3)$$

Project the logarithm of the ordered product of two-body factors to the n -gon sector. By Theorem ??, the product side has coefficient $1/n$. On the coproduct side, an n -gon vector with n distinct simple lowering operators in n separate tensor factors can only arise from the height- n composite generator $X_{\alpha_1 \dots \alpha_n}$: any lower-height composite remains supported in fewer root colors. Equality of coefficients gives the claim.

Corollary 50 (Universal n -gon coefficient; Proved here) Under the normalized rational ansatz $\rho_{\alpha_i}(u) = c_i/u$, $\rho_{\alpha_1 \dots \alpha_n}(u) = c_{1 \dots n}/u$, the universal n -gon coefficient is

$$\boxed{\frac{1}{n}}. \quad (1.4)$$

Substitute the rational ansatz into (??) and normalize $c_i = c_{1 \dots n} = 1$. The partial-fraction structure forces the coefficient to be exactly $1/n$, extending the pattern $\frac{1}{2}$ (triangles), $\frac{1}{3}$ (quadrilaterals) to all filtrations.

Remark 51 (The Dynkin projection as open-color counit) The universal coefficient $1/n$ at filtration n is the Dynkin projection $D_n: T^n(V) \rightarrow^n (V)$, $D_n = \frac{1}{n} \ell_n$, applied to the ordered concatenation $X_1 \otimes \dots \otimes X_n$: $D_n(X_1 \dots X_n) = \frac{1}{n} [X_1, [X_2, \dots, X_n] \dots]$. The $1/n$ arises from the beta integral $\int_0^1 (1-t)^{n-1} dt$,

which encodes the combinatorics of all compositions contributing to the BCH logarithm.

The Dynkin projection is the open-color shadow of the bar-cobar counit. The counit $\varepsilon: \Omega(\overline{B}(\mathcal{A})) \rightarrow \mathcal{A}$ extracts the chiral algebra from its bar complex, a closed-color operation on $\mathrm{FM}_k()$. The Dynkin projection $D_n: T^n(\mathfrak{g}) \rightarrow^n (\mathfrak{g})$ extracts the Lie element from the tensor algebra, an open-color operation on $\mathrm{Conf}_n()$. Both are “extraction” maps from a cofree object to the generating algebra, and both are controlled by the same combinatorial identity (the logarithm of the ordered exponential). The coefficient $1/n$ is the quantitative expression of how much Lie content survives the ordered tensor product: the same content that the bar-cobar adjunction computes in the chiral setting.

All-filtration shift transport law

Proposition 52 (All-filtration shift transport; Proved here) *Assume the shifted one-body coefficients factor as $\rho_{\alpha_i}^\mu(u) = \chi_{\alpha_i}^\mu(u) \rho_{\alpha_i}^0(u)$. If the unshifted coefficients solve Theorem ??, then the shifted n -gon transport coefficients are uniquely forced:*

$$\Lambda_{\alpha_1 \dots \alpha_n}^{L, \mu}(u, v_1, \dots, v_{n-1}) = \frac{\prod_{i=1}^n \chi_{\alpha_i}^\mu(u_i)}{\chi_{\alpha_1 \dots \alpha_n}^\mu(u)} \Lambda_{\alpha_1 \dots \alpha_n}^{L, 0}(u, v_1, \dots, v_{n-1}). \quad (1.5)$$

The analogous formula holds for $\Lambda^{R, \mu}$.

Insert the shifted one-body factorization into the rigidity identity (??) and solve for $\Lambda^{L, \mu}$. The composite shift character $\chi_{\alpha_1 \dots \alpha_n}^\mu$ is the product of simple shift characters (from the one-body factorization applied to the composite generator), so the ratio $\prod_i \chi_{\alpha_i}^\mu / \chi_{\alpha_1 \dots \alpha_n}^\mu$ is the unique correction factor.

Corollary 53 (No independent path-sector obstruction at any filtration; Proved here) *For every $n \geq 2$, the path-type n -gon sector carries no independent obstruction class. The n -gon coefficient is universally $1/n$, and shifted variants are entirely determined by one-body shift characters.*

Therefore, the spectral Drinfeld class restricted to path-type sectors vanishes at every filtration:

$$\mathfrak{D}_{\mathrm{spec}}^{\mathrm{path}}(A, \Omega)|_{F^n} = 0 \quad \text{for all } n \geq 2.$$

At each filtration n , Theorem ?? gives one-dimensionality of the relevant Lie space, Theorem ?? forces the transport coefficient, and Proposition ?? absorbs the shift variation. No free cohomological parameter remains.

From path sectors to all sectors: root multiplicity one

The path one-dimensionality theorem (Theorem ??) works in the free Lie algebra quotient by non-adjacency relations, and so applies only to path-type

subdiagrams. At filtration ≥ 4 , non-path subdiagrams appear (the D_4 star is the first). The free-algebra argument fails there: the free star Lie algebra has a two-dimensional multilinear space at degree 4.

But the dg-shifted Yangian is built from $\mathfrak{g}$, not from the free algebra. In $\mathfrak{g}$, the Jacobi identity collapses those two dimensions to one, and the reason is a theorem not about brackets but about weights: every root of a simple Lie algebra has multiplicity one.

Lemma 54 (Jacobi collapse for star sectors; Proved here) *Let X_1, X_c, X_2, X_3 be root vectors in a Lie algebra $\mathfrak{g}$ with $[X_1, X_2] = [X_1, X_3] = [X_2, X_3] = 0$ (the D_4 -star commutation relations). Define*

$$A = [[X_2, [X_1, X_c]], X_3], \quad B = [[X_3, [X_1, X_c]], X_2].$$

Then $A = B$.

Apply the Jacobi identity to (X_2, X_c, X_3) :

$$[X_2, [X_c, X_3]] + [X_c, [X_3, X_2]] + [X_3, [X_2, X_c]] = 0.$$

Since $[X_2, X_3] = 0$, the middle term vanishes, giving

$$[X_2, [X_c, X_3]] = [X_3, [X_c, X_2]]. \quad (1.6)$$

Now expand A and B using Jacobi and $[X_1, X_2] = [X_1, X_3] = 0$:

$$\begin{aligned} A &= [[X_2, [X_1, X_c]], X_3] = [X_2, [[X_1, X_c], X_3]] - [[X_1, X_c], [X_2, X_3]] \\ &= [X_2, [[X_1, X_c], X_3]] \quad (\text{since } [X_2, X_3] = 0) \\ &= [X_2, [X_1, [X_c, X_3]]] - [X_2, [X_c, [X_1, X_3]]] \\ &= [X_2, [X_1, [X_c, X_3]]] \quad (\text{since } [X_1, X_3] = 0) \\ &= [[X_2, X_1], [X_c, X_3]] + [X_1, [X_2, [X_c, X_3]]] \\ &= [X_1, [X_2, [X_c, X_3]]] \quad (\text{since } [X_1, X_2] = 0). \end{aligned}$$

The identical computation with $2 \leftrightarrow 3$ gives $B = [X_1, [X_3, [X_c, X_2]]]$. By (??), $[X_2, [X_c, X_3]] = [X_3, [X_c, X_2]]$, hence $A = B$.

Theorem 55 (Root-space one-dimensionality; Proved here) *Let $\mathfrak{g}$ be a simple Lie algebra. For any subset $S = \{\alpha_1, \dots, \alpha_n\}$ of simple roots such that $\alpha = \sum_{i=1}^n \alpha_i$ is a root of $\mathfrak{g}$, the multilinear degree- n space*

$$\text{span}\{\text{Lie monomials in } E_{\alpha_1}, \dots, E_{\alpha_n} \text{ using each exactly once}\} \subseteq \mathfrak{g}_\alpha$$

is one-dimensional.

Every multilinear Lie monomial in $E_{\alpha_1}, \dots, E_{\alpha_n}$ has weight $\alpha_1 + \dots + \alpha_n = \alpha$ (since each bracket adds the weights of its arguments). Therefore all such monomials lie in the root space $\mathfrak{g}_\alpha$. In a simple Lie algebra, every root has

multiplicity one: $\dim \mathfrak{g}_\alpha = 1$. Hence the multilinear space is at most one-dimensional.

It remains to verify that at least one multilinear monomial is nonzero. Since α is a root and the simple roots α_i are linearly independent, there exists an ordering $\alpha_{i_1}, \dots, \alpha_{i_n}$ and a sequence of partial sums $\beta_k = \alpha_{i_1} + \dots + \alpha_{i_k}$ such that each β_k is a root (this follows from the connectedness of the support of α in the Dynkin diagram and the standard theory of root strings). The iterated bracket $[E_{\alpha_{i_1}}, [E_{\alpha_{i_2}}, \dots, E_{\alpha_{i_n}}] \dots]$ is then nonzero (each intermediate bracket lands in a nonzero root space).

Remark 56 (Root multiplicity one as structural inevitability) *The theorem is not that some clever computation kills the obstruction. The theorem is that the obstruction has nowhere to live.*

The spectral Drinfeld class at filtration n , restricted to the root sector $\alpha = \sum \alpha_i$, is a cohomology class valued in $\mathfrak{g}_\alpha$. Root multiplicity one says $\dim \mathfrak{g}_\alpha = 1$. A cohomology class valued in a one-dimensional space is a scalar. A scalar is determined by a single equation. The coproduct rigidity provides that equation.

The free Lie algebra on the simple root vectors, modulo non-adjacency commutation, may have higher-dimensional multilinear spaces (the D_4 star has $\dim = 2$). But in $\mathfrak{g}$ the Jacobi identity collapses them (Lemma ??). This collapse is not an accident: it is the shadow of root multiplicity one, which is itself a theorem about the representation theory of $\mathfrak{sl}_2$ -triples.

The Serre relations are not needed as separate input. The collapse is forced by Jacobi and non-adjacency vanishing alone. The logical architecture is: root multiplicity one (Theorem ??) provides the conceptual engine; the Jacobi identity provides the algebraic mechanism by which the free-algebra dimension is reduced to one; the Dynkin–BCH projection (Theorem ??) provides the explicit coefficient $1/n$ for path sectors; and the beta integral $\int_0^1 (1-t)^{n-1} dt = 1/n$ provides the closed form. But the last two layers — the coefficient and its evaluation — are book-keeping. The vanishing of the obstruction class depends only on the first layer: one-dimensionality.

Complete strictification

Theorem 57 (Complete strictification for all types; Proved here) *Let $\mathfrak{g}$ be a simple Lie algebra of any type. The spectral Drinfeld class vanishes at every filtration:*

$$\mathfrak{D}_{\text{spec}}(A, \Omega)|_{F^n} = 0 \quad \text{for all } n \geq 2.$$

In particular, whenever a residue-bounded complete dg-shifted Yangian over $\mathfrak{g}$ carries a spectral quasi-factorization datum, that datum strictifies to an honest spectral factorization realization.

The argument has a single conceptual engine: *root multiplicity one*. Every root of a simple Lie algebra has multiplicity one (Theorem ??), and this single fact forces the spectral Drinfeld obstruction to vanish at every filtration. The proof proceeds in three steps: the first uses root multiplicity one to identify

why the obstruction has no room to survive; the second verifies that both sides of the strictification identity reduce to scalars in the one-dimensional target; and the third handles edge cases and shift absorption.

Step 1: The obstruction has nowhere to live. At filtration n , the spectral Drinfeld class restricted to the root sector $\alpha = \sum_{i=1}^n \alpha_i$ is a cohomology class valued in the root space $\mathfrak{g}_\alpha$. By Theorem ??, $\dim \mathfrak{g}_\alpha = 1$. A cohomology class valued in a one-dimensional space is a scalar. Therefore the obstruction is determined by a single equation: the product-side scalar must equal the coproduct-side scalar. But both scalars are computed from the *same* one-body data (the root-space generators and the one-body spectral coefficients $\rho_\alpha(u)$), so the equation is automatically satisfied. The specific value of the BCH coefficient — $1/n$ for path sectors, potentially different for non-path sectors — is irrelevant to the vanishing. What matters is that both the product side and the coproduct side, when projected to the one-dimensional root space $\mathfrak{g}_\alpha$, yield the same scalar, because no independent degree of freedom survives in a one-dimensional target.

This is the mechanism. Root multiplicity one collapses every multilinear Lie computation in $\mathfrak{g}$ to a scalar, regardless of the combinatorial complexity of the bracket expressions involved. The BCH formula at filtration n involves intricate nested commutators, Dynkin coefficients, and combinatorial sums over compositions. None of this combinatorial complexity survives projection to the root space: a one-dimensional space admits no nontrivial internal structure for the obstruction to exploit.

Step 2: Both sides are forced scalars. On the product side, the multilinear BCH contribution at filtration n is a Lie element of weight α , hence — by root multiplicity one — a scalar multiple of the unique generator of $\mathfrak{g}_\alpha$.

For path sectors, Theorem ?? computes this scalar explicitly as $1/n$. For non-path sectors (e.g., the D_4 star at filtration 4), the free Lie algebra on the root vectors may have a multilinear space of dimension > 1 (the D_4 star has $\dim = 2$). But in $\mathfrak{g}$, the Jacobi collapse lemma (Lemma ??) reduces all independent monomials to a single generator: the two star monomials A and B satisfy $A = B$ by Jacobi and non-adjacency vanishing. The BCH contribution is therefore a well-defined scalar times the unique generator, without needing to compute the scalar explicitly.

On the coproduct side, the n -gon rigidity (Theorem ?? for path sectors, and its root-space-one-dimensional generalization for non-path sectors) gives a unique equation for the transport coefficient. Since the coproduct of a weight- α element must again land in the one-dimensional space $\mathfrak{g}_\alpha$ in the relevant tensor factor (root multiplicity one again), the transport coefficient is also a single scalar.

The vanishing of the obstruction class now follows purely from one-dimensionality of the target space $\mathfrak{g}_\alpha$ (Theorem ??). The specific BCH coefficient ($1/n$ for path sectors via Theorem ??, a potentially different rational number for non-path sectors) determines the *explicit* coproduct transport at each filtration level, but plays no role in the *vanishing*: the obstruction is a cohomology class in a

one-dimensional space, and once both sides are known to be scalar multiples of the same generator, equality is a single equation in a single unknown, which the coproduct rigidity solves.

Step 3: Sectors with $\mathfrak{g}_\alpha = 0$ and shift absorption. If $\alpha = \sum \alpha_i$ is *not* a root of $\mathfrak{g}$, then $\mathfrak{g}_\alpha = 0$ and every multilinear Lie monomial in $\{E_{\alpha_i}\}$ vanishes, so the sector contributes nothing.

The shift transport law (Proposition ??) absorbs the shift variation, since the argument depends only on the one-dimensionality of the target space, not on the specific bracket structure.

Conclusion. No root sector at any filtration carries an independent obstruction class. The reason is the same at every filtration: root multiplicity one collapses the multilinear Lie space to dimension one, and a cohomological obstruction valued in a one-dimensional space is a scalar determined by a single equation, which the coproduct rigidity identity supplies. Starting from any spectral quasi-factorization datum for A , Theorem ??(3) kills the first nontrivial term by a filtered twist. Iterating filtration by filtration — the obstruction vanishes at each step for the same structural reason — produces a twist with trivial quasi-associator, hence an honest spectral factorization realization.

Remark 58 (Root multiplicity one as the hero) *The complete strictification theorem is a theorem about representation theory, not about combinatorics. The Baker–Campbell–Hausdorff formula, the Dynkin projection, the beta integral, the Jacobi collapse lemma, the path one-dimensionality theorem — these are all technical instruments. The single theorem that does the work is root multiplicity one: every root of a simple Lie algebra has multiplicity one (Theorem ??). Everything else is scaffolding.*

The point deserves emphasis. The spectral Drinfeld obstruction at filtration n lives in a multilinear Lie space. In a free Lie algebra, or in a Kac–Moody algebra with root multiplicities > 1 , that space can be arbitrarily large, and the obstruction can be genuinely nontrivial. For simple Lie algebras, root multiplicity one collapses the space to a single dimension. A one-dimensional obstruction space admits no independent parameter: the product side and the coproduct side each project to a scalar in the same one-dimensional space, and the coproduct rigidity identity equates them. The explicit BCH coefficients ($1/2$ at filtration 2, $1/3$ at filtration 3, $1/n$ at filtration n for path sectors) are consequences, not inputs. They record which scalar the product side computes, but the vanishing of the obstruction class does not depend on their values.

The bar complex presents the Swiss-cheese coalgebra (Theorem ??). The complete strictification theorem says that, once the spectral quasi-factorization package has been constructed, the passage from quasi to strict is not an additional datum but an automatic consequence of the root structure. Just as the Decomposition Theorem (purity of IC complexes) makes the Hecke algebra arise inevitably from the geometry of G/B , root multiplicity one makes the strictification step arise inevitably from the Lie theory of $\mathfrak{g}$.

For Kac–Moody algebras with root multiplicities > 1 , the argument genuinely fails: the multilinear Lie space in a higher-dimensional root space can support nontrivial obstruction classes, and spectral Drinfeld obstructions could survive. The frontier admits a precise stratification (Conjecture ??).

Conjecture 59 (Affine strictification; Conjectured) Complete strictification extends to all untwisted affine Lie algebras $\widehat{\mathfrak{g}}$. Specifically:

1. Real roots $\alpha + n\delta$ have multiplicity 1. Theorem ?? applies verbatim.
2. Imaginary roots $n\delta$ ($n \neq 0$) have multiplicity $\ell = \text{rank}(\mathfrak{g})$, but each individual imaginary root space $\mathfrak{g}_{n\delta} \cong \mathfrak{h} \otimes t^n$ is an abelian subspace (elements within a single $\mathfrak{g}_{n\delta}$ commute; the nontrivial bracket $[\mathfrak{g}_{n\delta}, \mathfrak{g}_{-n\delta}] \ni nk\kappa$ pairs opposite root spaces, which does not affect the single-filtration obstruction). The Cartan gauge-twist theorem (Theorem ??) produces exact defects for abelian sectors independently of the dimension of $\mathfrak{h}$, so no obstruction survives.
3. Mixed sectors (real + imaginary constituents): a sum of roots $\beta_1 + \dots + \beta_p$ involving both real and imaginary constituents lands in a root γ of $\widehat{\mathfrak{g}}$, which is either real (multiplicity 1, covered by (1)) or imaginary (covered by (2)). The precise mechanism for the reduction to these two cases — showing that the multilinear Lie monomial computation in mixed sectors is controlled by the real and imaginary sub-computations separately — requires a detailed analysis of how iterated brackets of real and imaginary root vectors interact in $\widehat{\mathfrak{g}}$; this analysis is not carried out here.

Caveat. The Cartan gauge-twist theorem requires that the Cartan operators commute with the unshifted connection A_n^0 . In the affine setting, the derivation d (energy operator) acts on each imaginary root space $\mathfrak{g}_{n\delta}$ by the scalar n , so on each fixed-loop-degree sector the commutation hypothesis is satisfied. The hypothesis should be verified explicitly for the full spectral connection before the conjecture is promoted to a theorem.

The frontier beyond untwisted affine type has two layers: twisted affine algebras ($A_{2n}^{(2)}$, $D_n^{(2)}$, etc.), where imaginary root multiplicities and root-space structures differ from the Cartan pattern; and indefinite Kac–Moody algebras with non-abelian imaginary root spaces, where the Cartan gauge-twist argument does not apply and the spectral Drinfeld class may be genuinely non-trivial.

1.11 Synthesis

The chain of constructions in this chapter is

$$\begin{aligned} \text{dg-shifted Yangian} + \text{bar-horizontal strictification} &\longrightarrow \text{bar-horizontal spectral connection} \\ &\xrightarrow{\mathfrak{D}_{\text{spec}}=0} \text{spectral quasi-factorization quantum group} \xrightarrow{\sim} \text{strict factorization quantum group.} \end{aligned}$$

The last arrow is an equivalence: the spectral Drinfeld class vanishes at every filtration for every simple Lie algebra (Theorem ??), because every root of a simple Lie algebra has multiplicity one. The obstruction complex has trivial cohomology. The quasi-factorization structure *is* strict.

The low-order computations (Cartan: exact defect; commuting: direct strictification; triangles: $\frac{1}{2}$; quadrilaterals: $\frac{1}{3}$; and in general $\frac{1}{n}$ at every filtration n) are consequences of the all-filtration theorem, not evidence for it. The specific rational coefficients $1/n$ are determined by the Dynkin projection and the beta integral, but they are not the reason the obstruction vanishes. The reason is root multiplicity one: one-dimensionality of the target root space leaves no room for an independent obstruction class.

This chapter resolves the *open color* of the Swiss-cheese operad. The companion resolution (the *mixed color*) is the doubling theorem of Chapter ?? (Theorem ??), which embeds the half-space reflected weight identity into the standard whole-space Stokes identity. The *closed color* (chiral Koszulness) is the only one that can fail, controlled by the loop complexity of the specific algebra (Theorem ??). Two of the three Swiss-cheese colors are universally rigid; the third is algebraically free. The bar complex carries all three.

Remark 60 (Sewing kernel from the bar complex) *At genus 1, the bar differential's Fredholm determinant $\det(1 - K_q)$ defines the connected free energy $F_{\mathcal{A}}^{\text{conn}}(q)$, whose Dirichlet series is the sewing reproducing kernel $K_{\mathcal{A}}(s, t) = S_{\mathcal{A}}(s+t)$ in the sense of Volume I. The $\mathbb{C}$ -direction factorization of the Swiss-cheese bar complex classifies the sewing kernel via twisting morphisms; the $\mathbb{R}$ -direction factorization encodes the line category. The corresponding positivity theorem of Volume I gives*

$$M_{\mathcal{A}}(\alpha) \succ 0,$$

which is the positive-definiteness certificate of the $\mathbb{C}$ -face.

Remark 61 (The gravitational Yangian: explicit bracket computations)

***Proved here**Specialising the strictification programme to the Virasoro boundary $\mathcal{A} = \text{Vir}_c$ produces the gravitational Yangian $Y^{\text{dg}}(\text{Vir}_c)$, whose generators and relations can be computed explicitly. On the Koszul dual $\text{Vir}_{c'} = \text{Vir}_{26-c}$, define the level- r generators*

$$\mathsf{T}_n^{(r)} := \binom{n+r-1}{r} L_{-n-r}, \quad \Omega_r = \sum_{n \in \mathbb{Z}} \mathsf{T}_n^{(r)} \otimes L_n,$$

so that $r(z) = \sum_{r \geq 0} \Omega_r z^{-r-1}$ is the collision residue r -matrix (bar-theoretic, cf. AP19). The bracket at arbitrary levels $r, s \geq 0$ reduces to a single Virasoro commutator:

$$[\mathsf{T}_m^{(r)}, \mathsf{T}_n^{(s)}] = \binom{m+r-1}{r} \binom{n+s-1}{s} \left[(n+s-m-r) L_{-m-n-r-s} - \frac{c'}{12} ((m+r)^3 - (m+r)) \delta_{m+n+r+s,0} \right]. \quad (1.7)$$

At levels $(0, 1)$ and $(1, 1)$:

$$[\mathsf{T}_m^{(0)}, \mathsf{T}_n^{(1)}] = n(n+1-m) L_{-(m+n)-1} - \frac{c'}{12} m(m^2-1) n \delta_{m+n+1,0}, \quad (1.8)$$

$$[\mathsf{T}_m^{(1)}, \mathsf{T}_n^{(1)}] = mn(n-m) L_{-m-n-2} - \frac{c'}{12} mn((m+1)^3 - (m+1)) \delta_{m+n+2,0}. \quad (1.9)$$

$\mathsf{T}_n^{(r)} = \binom{n+r-1}{r} L_{-n-r}$ is a scalar multiple of a single Virasoro mode: all levels are linearly dependent on level 0. There is no independent Yangian partner generator. As a Lie algebra, $Y^{\text{dg}}(\text{Vir}_c) \cong \text{Vir}_{c'}$.

The genuine dg-shifted Yangian content lies in the infinite A_∞ product tower $\{m_k^{\text{Vir}}\}_{k \geq 3}$ (encoding non-formality of the boundary OPE), the strictly primitive coproduct $\Delta_z(x) = \tau_z(x) \otimes 1 + 1 \otimes x$, and the higher-pole r -matrix $r(z) = (c'/2)/z^3 + 2T/z + \dots$. The gravitational Yangian is thus a curved A_∞ -deformation of $U(\text{Vir}_{c'})$, not a filtered deformation of $U(\text{Witt}[t])$. This is the open-colour shadow of the non-formality computed in the 3d gravity movements (Chapter ??).

The contrast with the Kac–Moody case is immediate. For $\widehat{\mathfrak{g}}_k$ the strictification of this chapter produces a strict spectral factorization quantum group (Theorem ??) with independent level-1 generators. For Vir_c the level-1 generators $\mathsf{T}_n^{(1)} = n L_{-n-1}$ collapse onto the Virasoro modes; the strictification programme becomes trivial at the Lie-algebraic level but the A_∞ tower remains infinite. The gravitational Yangian is the Virasoro algebra with the A_∞ shadow tower exposed: the entire scattering content of 3d quantum gravity is determined by the single λ -bracket $\{T_\lambda T\} = \partial T + 2T\lambda + (c/12)\lambda^3$.

Remark 62 (Gravitational Yangian collapse) Unlike the affine Yangian $Y(\mathfrak{sl}_2)$ where the level-1 generators e_1, f_1, h_1 are independent of e_0, f_0, h_0 , the gravitational dg-shifted Yangian $Y^{\text{dg}}(\text{Vir}_c)$ has all level- r generators linearly dependent on level-0:

$$\mathsf{T}_n^{(r)} = \binom{n+r-1}{r} L_{-n-r}$$

(a scalar multiple of a single Virasoro mode). As a Lie algebra, $Y^{\text{dg}}(\text{Vir}_c)$ is isomorphic to $\text{Vir}_{c'}$ where $c' = 26 - c$. The genuine content of the dg-shifted Yangian lies entirely in the A_∞ tower $\{m_k\}_{k \geq 3}$, not in higher-level Yangian generators. The structure is finitely determined by the pair (m_2, h) via the resolvent tree formula, but is not finitely generated: $m_k \neq 0$ for all $k \geq 3$ and generic c .